Digital financial inclusion, institutional quality and income inequality

Evidence from 36 developing economies, 2017-2022. Research summary for PhD admissions.

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Summary

Mobile money is often said to narrow income gaps in developing countries. Our team asked whether that depends on how well a country is governed. We built a panel of 36 developing economies (143 country-year observations, 2017-2022), measured mobile-money access with a principal-component index, measured governance with the six World Bank governance indicators, and estimated two-way fixed-effects models on the SWIID disposable-income Gini.

The first draft showed a borderline-significant interaction (p about 0.05). An independent replication of the R code in Python then exposed two errors: standard errors that ignored the degrees of freedom used by the fixed effects, and a bank-branch control built from raw counts instead of branches per 100,000 adults. After correcting both, the interaction keeps its negative sign in every robustness run but is no longer statistically significant at conventional levels (p about 0.15). We rewrote the paper around what the data can support. Its main finding is methodological: when the six governance indicators are entered separately, the interaction estimates split in sign and the standard errors balloon, while a single composite index gives one stable estimate.

Panel	143 country-years, 36 economies, 2017-2022 (unbalanced)
Outcome	SWIID disposable-income Gini
Predictor	Mobile-money index (PC1 of accounts, agents, transactions; 78.5% of variance)
Moderator	Institutional quality index (PC1 of six WGI dimensions; 77.7% of variance)
Estimator	Two-way fixed effects, standard errors clustered by country
Status	Preprint on Research Square; manuscript prepared for Empirical Economics (not yet submitted)

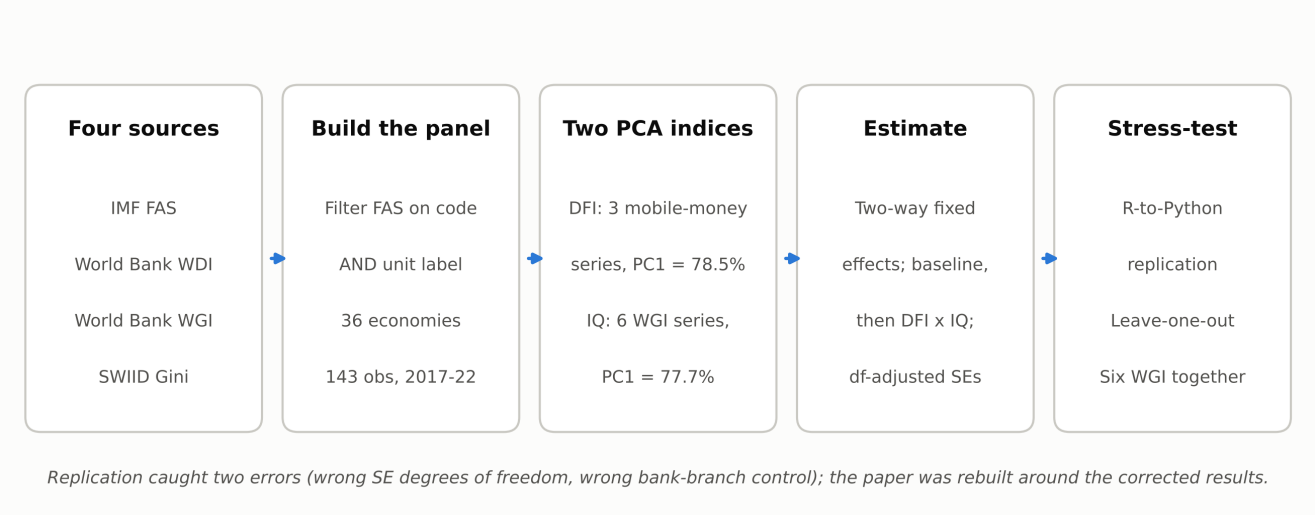


Figure 1. Workflow from raw downloads to stress-tested estimates.

Question and design

The paper's question is whether digital financial inclusion lowers income inequality, and whether the effect is larger where institutions are stronger. The theory rests on four channels (credit access, savings and consumption smoothing, payments and remittances, and entrepreneurship). Weak governance can throttle each of them, for example through poorly regulated platforms or low trust.

Data. Mobile-money indicators come from the IMF Financial Access Survey, governance from the World Bank WGI, controls (log GDP per capita, trade, inflation, urbanisation, remittances, bank branches) from WDI, and the Gini from SWIID. SWIID replaced the World Bank Gini because the latter left only 72 usable observations. The 2017-2022 window covers the period of rapid mobile-money growth.

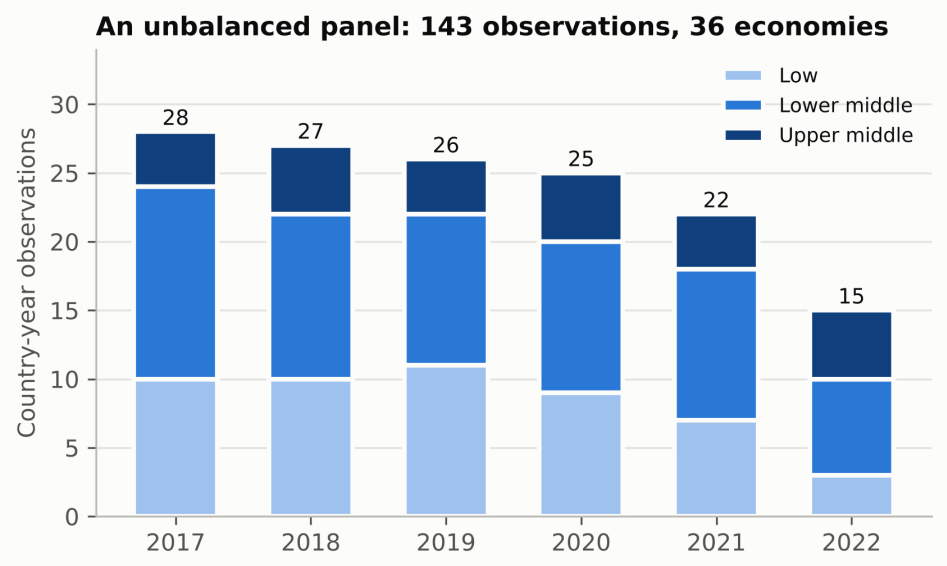


Figure 2. Observations by year and World Bank income group. Fewer countries report in later years, which is why the panel is unbalanced.

A data trap worth naming. The IMF file reports the same indicator under several units. Filtering on the indicator code alone mixed raw counts with per-capita values, so the first version of the index tracked country size rather than inclusion. Filtering on indicator code and unit label together fixed it, and a spot check (Kenya 2020: about 2,082 accounts per 1,000 adults) now guards against it returning.

Findings

1. No average effect. The baseline coefficient on the mobile-money index is -0.07 Gini points ($p = 0.68$). The research design anticipated this: if the effect depends on institutions, the average can sit near zero.

2. A negative interaction that stops short of significance. The interaction of mobile money with institutional quality is -0.077 (standard error 0.053, $p = 0.15$; within R-squared 0.12). Dropping any single country never changes the sign (36 of 36 runs, range -0.104 to -0.058), but only 2 of 36 runs reach p below 0.10. The sign is stable, the evidence is weak.

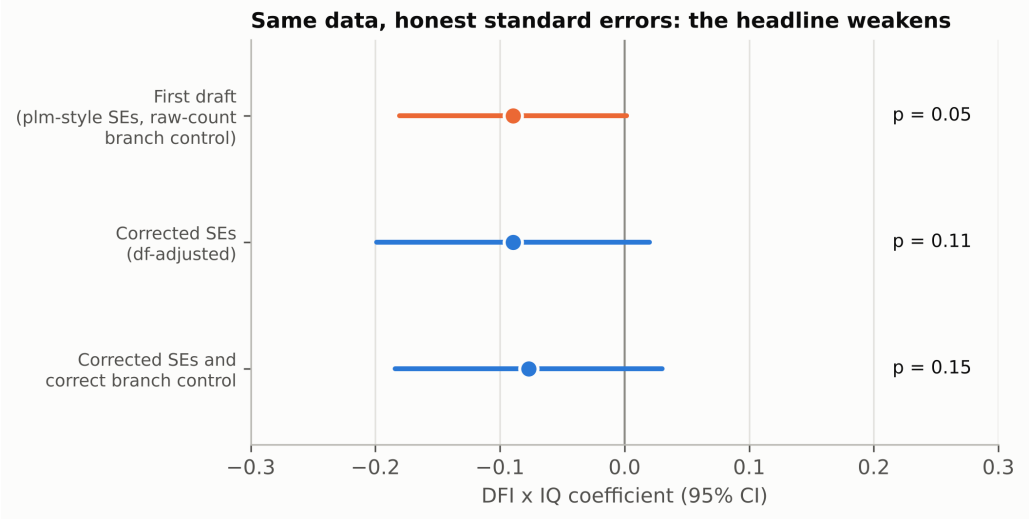


Figure 3. The same interaction under three treatments. Replacing plm-style standard errors with degrees-of-freedom-adjusted ones, and then fixing the branch control, moves p from about 0.05 to 0.15. The first bar approximates the first-draft setup.

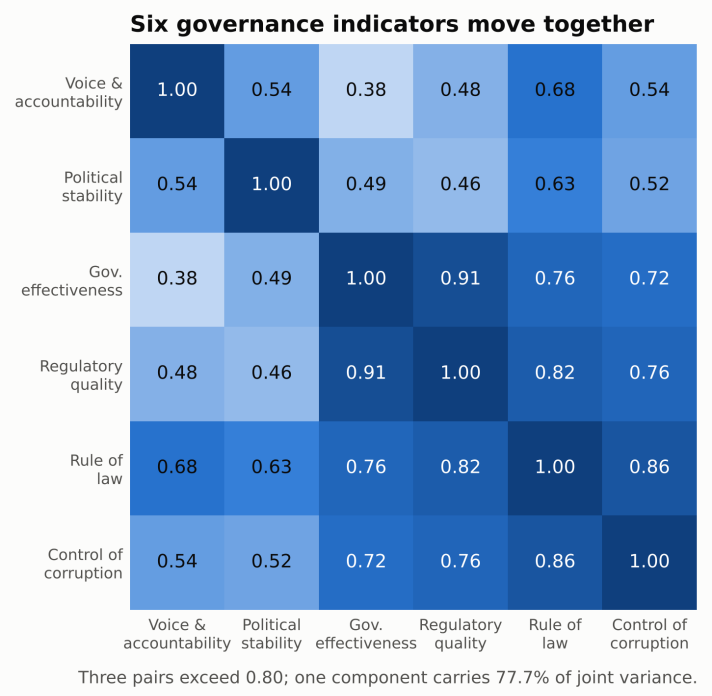


Figure 4. Correlations among the six governance indicators. Government effectiveness and regulatory quality correlate at 0.91, rule of law and corruption control at 0.86. Variance inflation factors run from 1.9 to 7.7 (mean 4.9), so the overlap is substantial but not extreme.

3. Separate governance dimensions give unstable answers. With all six interactions in one model, four come out positive and two negative, and none is significant at 5%. A single composite index gives one coefficient with one sign. This is the contrast the paper is built around: it suggests that contradictory institutional results in earlier work may reflect collinearity in how governance is entered, rather than real differences between dimensions.

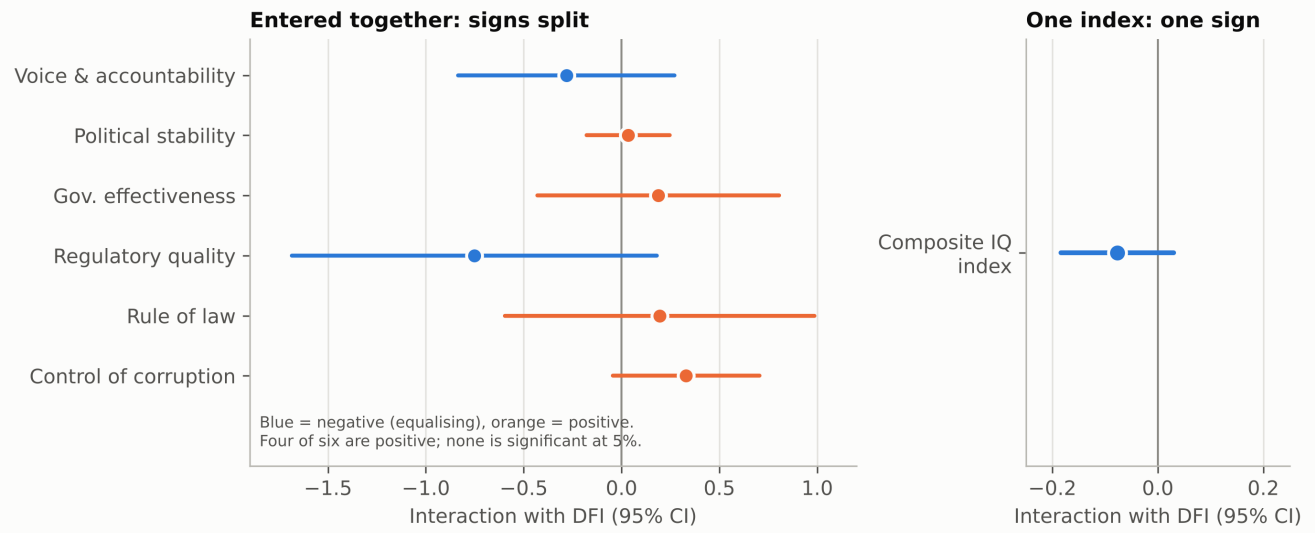


Figure 5. Left: six governance interactions estimated together, with 95% intervals. Right: the composite index interaction. The two panels use different horizontal scales because the variables are measured differently.

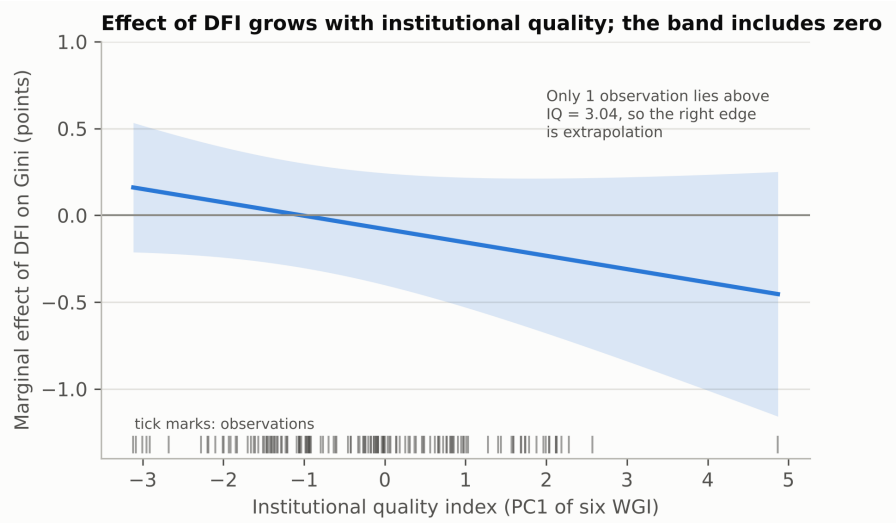


Figure 6. Marginal effect of mobile money on the Gini across the institutional-quality range. The point estimate falls as institutions improve, but the 95% band includes zero everywhere. Only one observation sits above 3.04, so the right edge is extrapolation. For that reason the paper does not report a governance threshold.

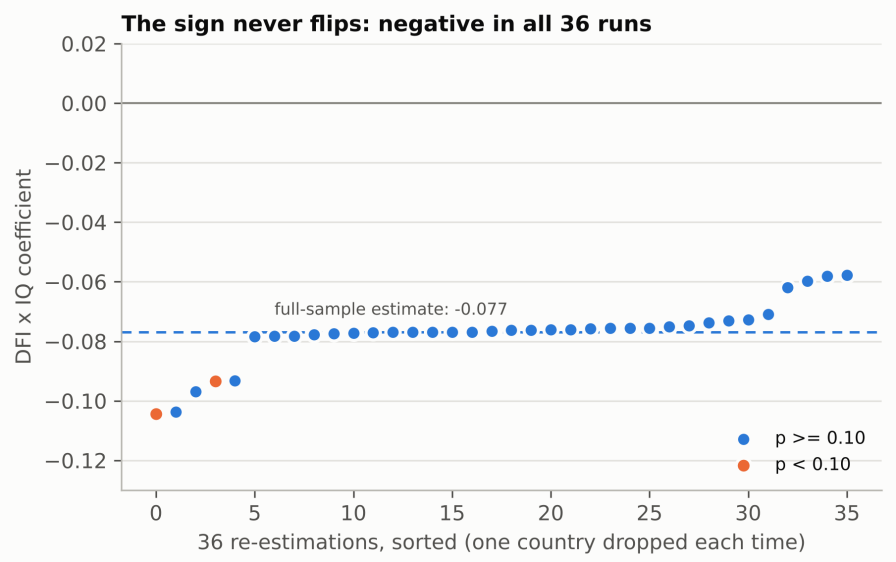


Figure 7. Interaction estimate after dropping each country in turn. All 36 estimates are negative and close to the full-sample value.

4. Income groups. Splitting the sample gives -0.075 (p = 0.41) for upper-middle-income countries (27 observations, 8 countries) and -0.053 (p = 0.45) for the rest (116 observations, 28 countries). Both point the same way and both

are underpowered, so we treat this as suggestive only.

Implications and limits

- **For policy.** The data do not show that mobile money narrows inequality on average. They are consistent with larger gains where governance is stronger, but this sample cannot confirm it. Pairing digital-finance rollouts with regulatory capacity is a hypothesis worth testing, not a finding.
- **For method.** Entering highly correlated governance indicators one by one, or all at once, can produce sign changes that look like substance. A composite index, or an explicit check of collinearity, should come first.
- **For inference.** With 36 clusters, a borderline p-value can move a long way once the degrees-of-freedom correction is applied. We report the corrected value.
- **Limits.** Short panel; unbalanced coverage; no formal treatment of endogeneity (reverse causation between inequality and adoption is plausible); observational design.

Next steps

- Extend the panel backwards and use regulatory events (for example agent-banking rules) for an event-study design, which speaks to causality better than fixed effects alone.
- Check inference with a wild cluster bootstrap, given few clusters.
- Test sensitivity to the choice of inequality measure and mobile-money indicators.

What this shows about how I work

The part of this project I value most is the verification step. Re-running the pipeline independently in a second language caught problems that looked plausible in the first. We then reported the weaker result instead of defending the first one. I want to bring the same habit, checking a result against an independent calculation before trusting it, to computational research in my PhD field. My background is in naval architecture and marine engineering (B.Sc., BUET, 2024), and I am now an MBA student at the Institute of Business Administration, University of Dhaka. The tools used here were R (plm) for estimation, Python for replication and figures, and Node.js to build the journal-formatted manuscript.

Authors. Md. Iftekharul Amin (supervising faculty, first author), Walid, Nasib (corresponding author), Alif Bin Hasan, Ittesaf Rahman. Data: IMF Financial Access Survey; World Bank WDI and WGI; SWIID. Estimates in this summary were recomputed from the project dataset with the corrected branch control and degrees-of-freedom-adjusted clustered standard errors; the manuscript tables are the authoritative version.